Assignment 2

2025-03-22

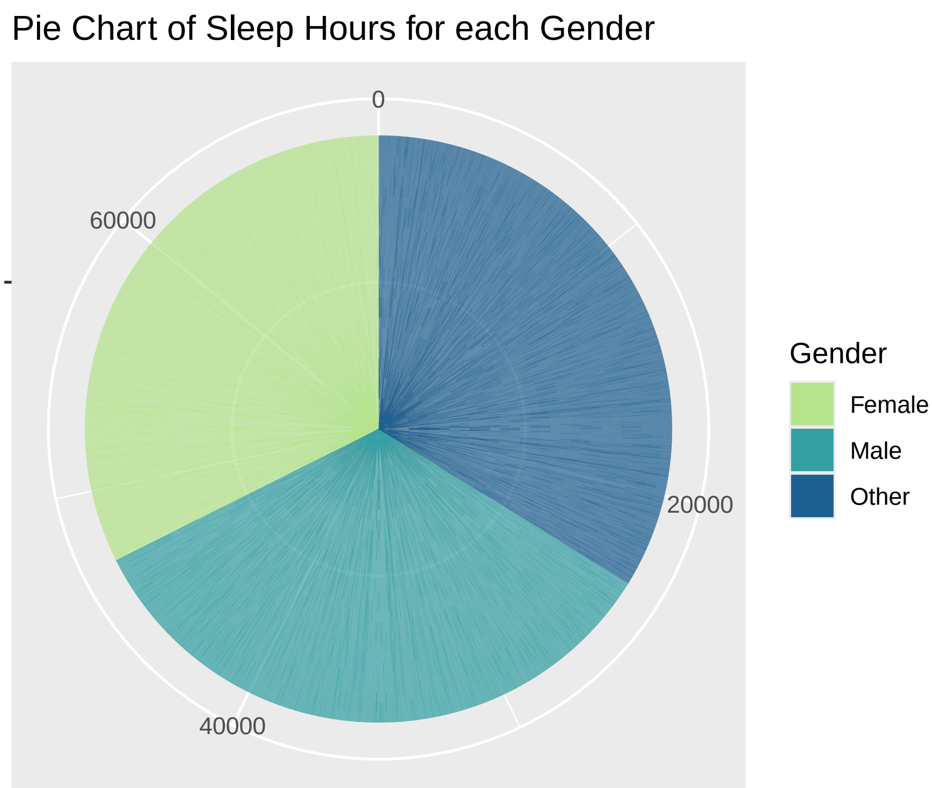
```r
library(ggplot2)
library(ggridges) #for ridgeline plot
library(ggforce) #for pie chart

data = read.csv("workout_fitness_tracker_data.csv")
head(data)
```

```
##   User.ID Age Gender Height..cm. Weight..kg. Workout.Type
## 1       1  39   Male         175          99      Cycling
## 2       2  36  Other         157         112       Cardio
## 3       3  25 Female         180          66         HIIT
## 4       4  56   Male         154          89      Cycling
## 5       5  53  Other         194          59     Strength
## 6       6  33   Male         162          81         Yoga
##   Workout.Duration..mins. Calories.Burned Heart.Rate..bpm. Steps.Taken
## 1                      79             384              112        8850
## 2                      73             612              168        2821
## 3                      27             540              133       18898
## 4                      39             672              118       14102
## 5                      56             410              170       16518
## 6                      11             301              142       10895
##   Distance..km. Workout.Intensity Sleep.Hours Water.Intake..liters.
## 1         14.44              High         8.2                   1.9
## 2          1.10              High         8.6                   1.9
## 3          7.28              High         9.8                   1.9
## 4          6.55            Medium         5.8                   1.9
## 5          3.17            Medium         7.3                   1.9
## 6          6.53               Low         4.2                   1.9
##   Daily.Calories.Intake Resting.Heart.Rate..bpm. VO2.Max Body.Fat....
## 1                  3195                       61    38.4         28.5
## 2                  2541                       73    38.4         28.5
## 3                  3362                       80    38.4         28.5
## 4                  2071                       65    38.4         28.5
## 5                  3298                       59    38.4         28.5
## 6                  2401                       69    38.4         28.5
##   Mood.Before.Workout Mood.After.Workout
## 1               Tired           Fatigued
## 2               Happy          Energized
## 3               Happy           Fatigued
## 4             Neutral            Neutral
## 5            Stressed          Energized
## 6               Happy            Neutral
```

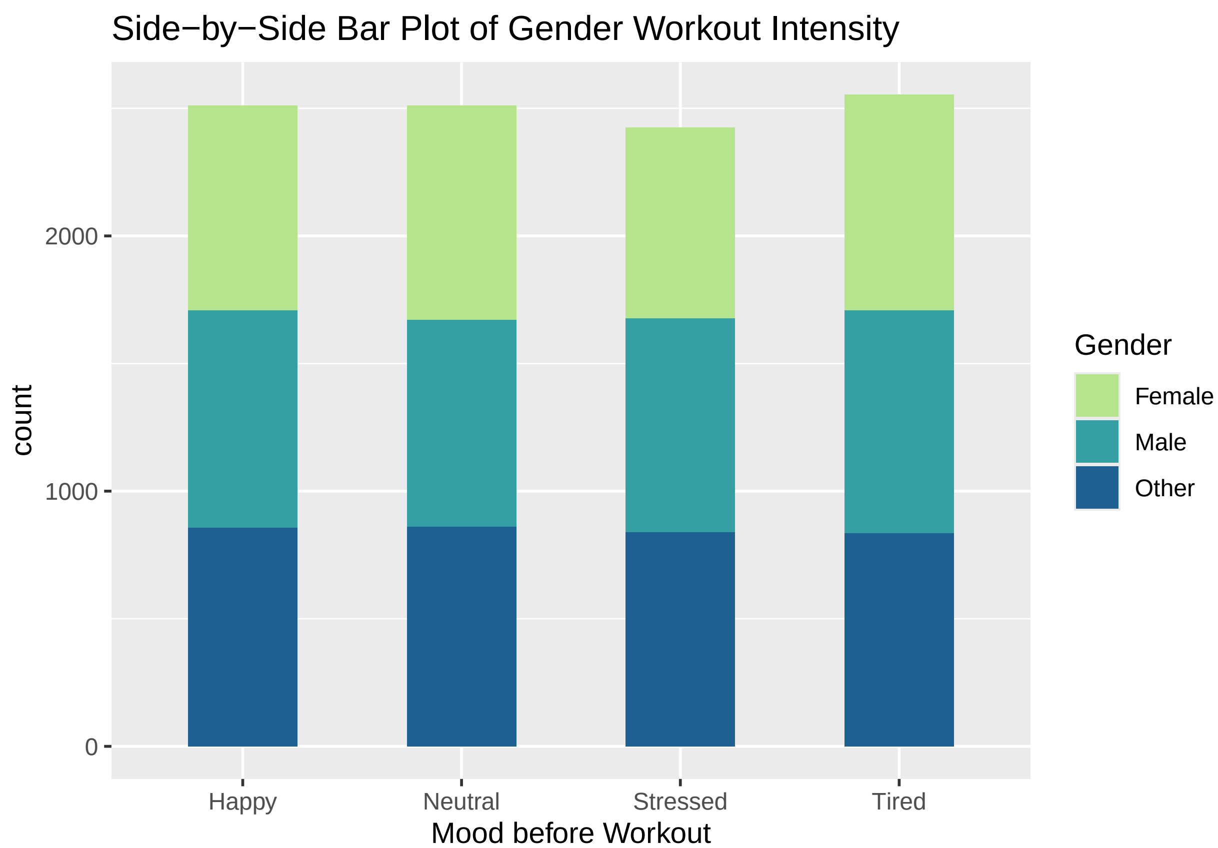
```r
#pie chart

ggplot(data, aes(x = "", y = Sleep.Hours, fill = Gender)) +
  geom_col(width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Pie Chart of Sleep Hours for each Gender", x = "", y = "") +
  scale_fill_manual(values = c("#B5E48C", "#34A0A4", "#1E6091"))
```




```r
#stacked bar plot

ggplot(data, aes(x = Mood.Before.Workout, fill = Gender)) +
     geom_bar(width=.5) +
     scale_fill_manual(values = c("#B5E48C", "#34A0A4", "#1E6091")) +
     labs(title = "Side-by-Side Bar Plot of Gender Workout Intensity", x = "Mood before Workout")
```

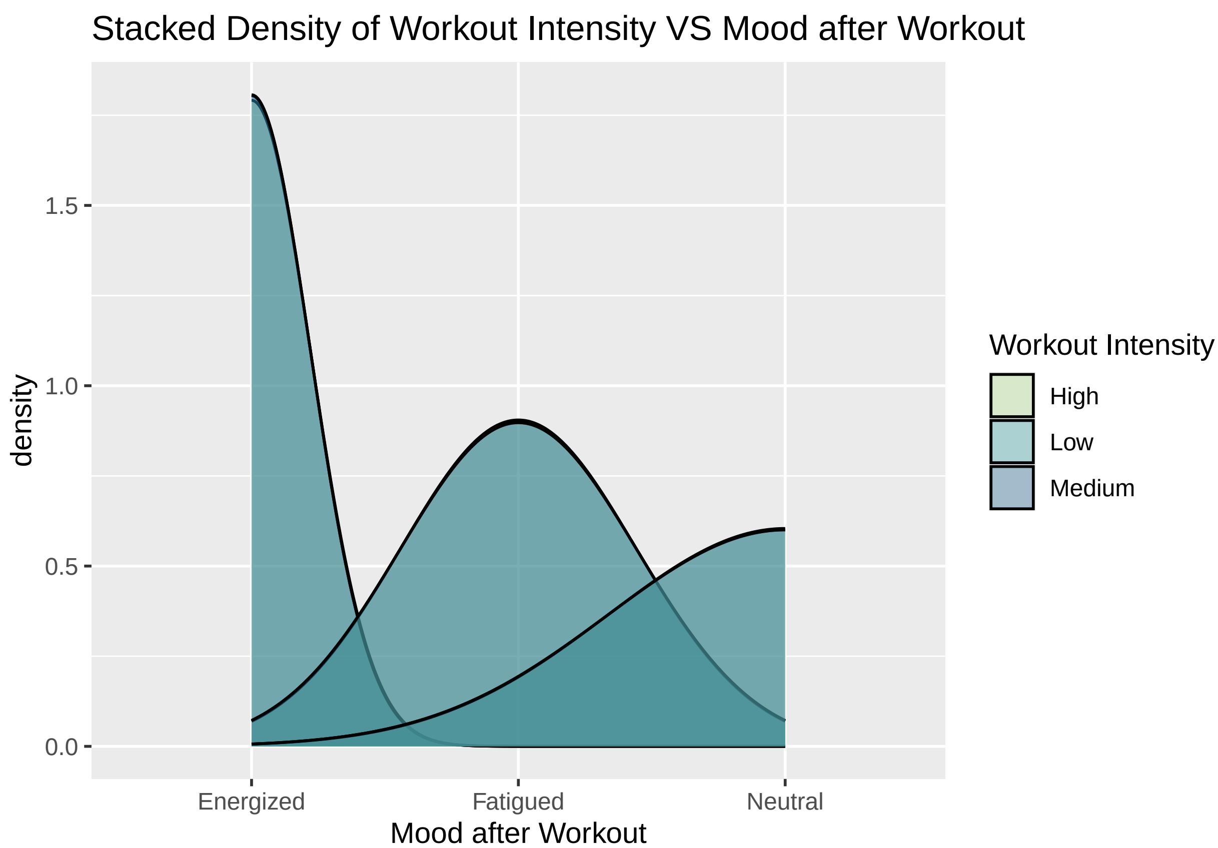



```r
#side-by-side bar plots

ggplot(data, aes(x = Workout.Intensity, fill = Gender)) +
     geom_bar(width=.5, position = "dodge") +
     scale_fill_manual(values = c("#B5E48C", "#34A0A4", "#1E6091")) +
     labs(title = "Side-by-Side Bar Plot of Gender Workout Intensity", x = "Intensity of Workout")
```

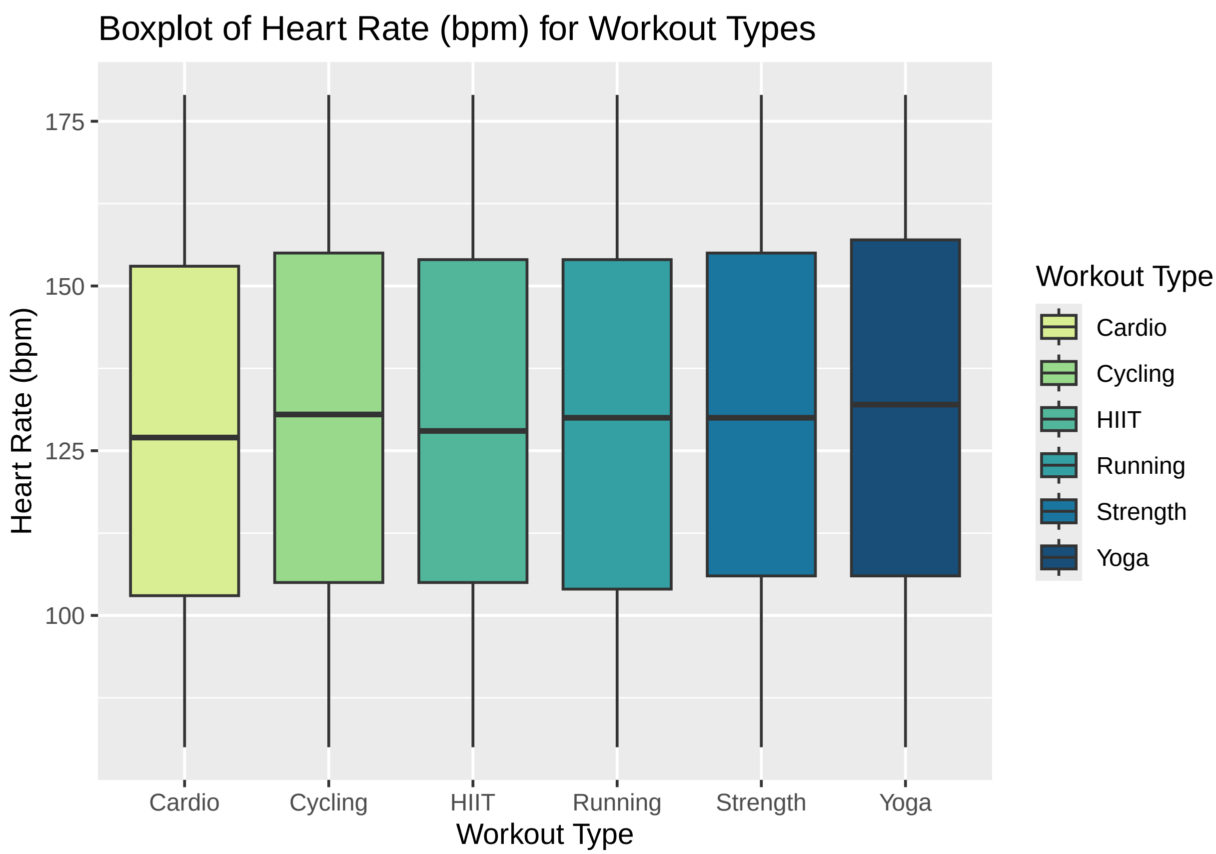

```r
#stacked densities

ggplot(data, aes(x = Mood.After.Workout, fill = Workout.Intensity)) +
  geom_density(alpha = 0.35) +
  scale_fill_manual(values = c("#B5E48C", "#34A0A4", "#1E6091")) +
  labs(title = "Stacked Density of Workout Intensity VS Mood after Workout", x = "Mood after Workout",
       fill = "Workout Intensity")
```




```r
#box plot

ggplot(data, aes(x = Workout.Type, y = Heart.Rate..bpm., fill = Workout.Type)) +
  geom_boxplot(outlier.colour = "red", outlier.shape = 16, outlier.size = 2) +
  scale_fill_manual(values = c("#D9ED92", "#99D98C", "#52B69A", "#34A0A4", "#1A759F", "#184E77")) +
  labs(title = "Boxplot of Heart Rate (bpm) for Workout Types", x = "Workout Type",
       y = "Heart Rate (bpm)", fill = "Workout Type")
```



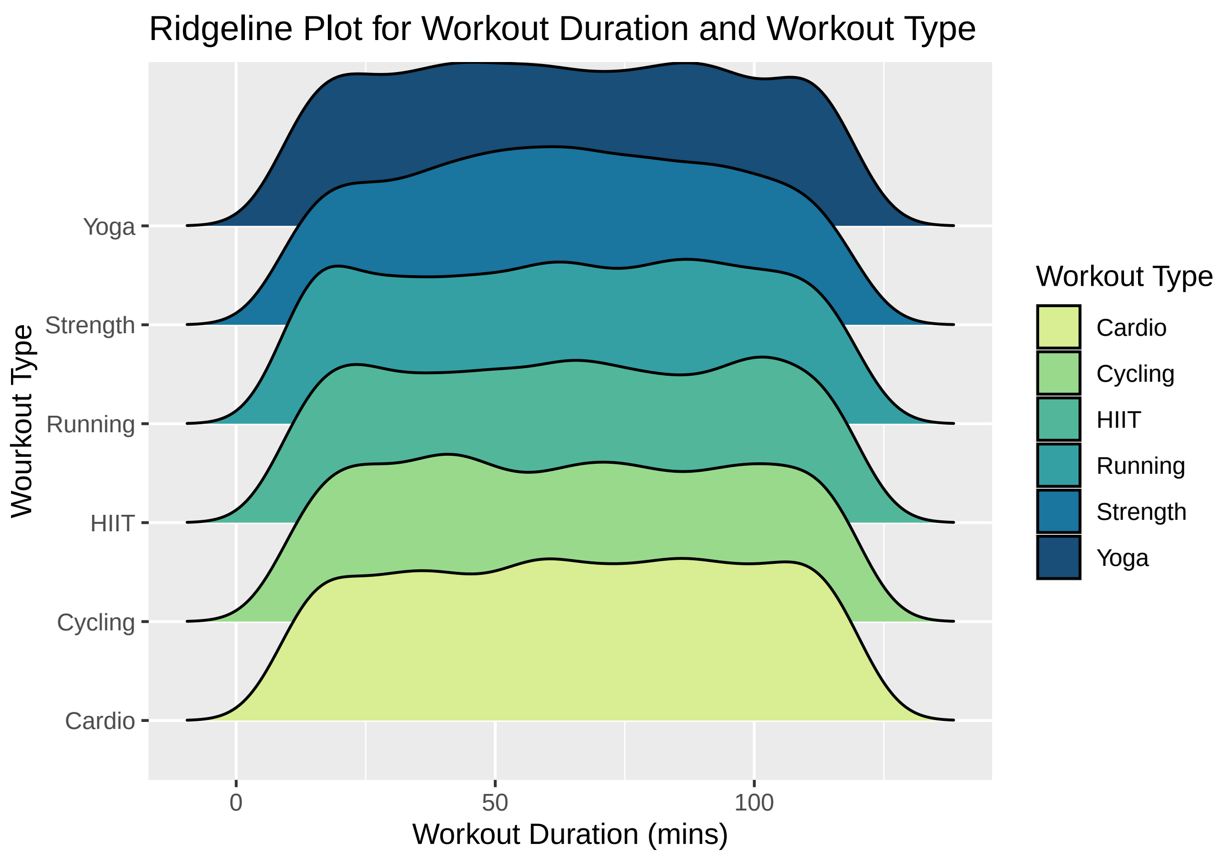

```
#violin plot

ggplot(data, aes(x = Gender, y = Steps.Taken, fill = Gender)) +
  geom_violin() +
  scale_fill_manual(values = c("#B5E48C", "#34A0A4", "#1E6091")) +
  labs(title = "Violin Plot for Steps Taken for Genders", y = "Steps Taken")
```


```r
#ridgeline plot

ggplot(data, aes(x = Workout.Duration..mins., y = Workout.Type, fill = Workout.Type)) +
  geom_density_ridges() +
  scale_fill_manual(values = c("#D9ED92", "#99D98C", "#52B69A", "#34A0A4", "#1A759F", "#184E77")) +
  labs(title = "Ridgeline Plot for Workout Duration and Workout Type", x = "Workout Duration (mins)", y
```

```
## Picking joint bandwidth of 6.49
```


Ridgeline Plot for Workout Duration and Workout Type
Wourkout Type
Yoga
Strength
Running
HIIT
Cycling
Cardio
0
50
100
Workout Duration (mins)
Workout Type
Cardio
Cycling
HIIT
Running
Strength
Yoga